ESPRIT – Data Engineering Project | 3ALINFO

SKY INSIGHT

Airline Loyalty & Passenger Satisfaction — End-to-End Data Engineering Platform

BI · Machine Learning · NLP · Computer Vision · SDG 9 · 12 · 13

PROJECT OVERVIEW

Business Context

An airline company seeks to harness its data assets **to boost passenger satisfaction, strengthen loyalty programs, and reduce its environmental footprint.**

This project delivers a full DE pipeline — from raw CSVs to insights, predictions, and sustainability reporting.

Datasets

Customer Loyalty History

CLV, loyalty cards, demographics

Customer Flight Activity

Flights, distance, points earned/redeemed

Passenger Satisfaction

24 service ratings + satisfaction label

Loyalty Data Dictionary

Schema reference

Calendar Table

Date dimension for DWH

DATA ARCHITECTURE

Sources

CSV files
Loyalty | Flights
Satisfaction

ETL/Staging

SSIS / Python
Cleaning
Transformation

Data Warehouse

SQL Server
Star Schema
Fact & Dims

BI Layer

Power BI
Dashboards
KPIs

AI/ML Layer

ML · NLP
Computer Vision
Predictions

External Data (SDG metrics, weather, carbon DB) feeds directly into the AI layer for enriched sustainability modelling

BUSINESS INTELLIGENCE OBJECTIVES

Loyalty Dashboard

Track CLV evolution, loyalty card tier distribution, enrollment trends, and churn signals by geography and segment.

Flight Performance Analytics

Monitor distance flown, points accumulation vs. redemption rate, peak travel months, and per-loyalty-tier revenue contribution.

Passenger Satisfaction KPIs

Drill down into 24 service dimensions — seat comfort, Wi-Fi, boarding — broken by class, travel type, and route.

Operational Efficiency

SDG 9 alignment: route utilisation, capacity fill rates, on-time performance, and load factor per fleet segment.

SDG 9: Industry Innovation and infrastructure

Environmental Reporting

SDG 12 & 13: fuel consumption per passenger-km, CO_2 intensity by route, emission trend over time, carbon offset tracking.

SDG 12: Responsible consumption and Production
SDG 13: Climate action carbon

Executive Summary Page

Single-page Power BI scorecard combining loyalty health, satisfaction index, ops efficiency, and carbon KPI in one view.

MACHINE LEARNING OBJECTIVES

Satisfaction Classifier

Binary classification (Satisfied vs. Neutral/Dissatisfied) using Random Forest / XGBoost on 24 service features. Target: AUC > 0.88.

Supervised · Classification

Loyalty Churn Predictor

Predict loyalty card cancellation using historical flight activity and CLV trends. Trigger early-intervention campaigns.

Supervised · Binary

Customer Segmentation

K-Means / DBSCAN clustering on loyalty & flight features to identify High-Value, At-Risk, Dormant, and Eco-Conscious segments.

Unsupervised · Clustering

Smart Route Optimisation (SDG 9)

Predict optimal flight load factor and recommend schedule adjustments to maximise utilisation and reduce per-seat emissions.

Regression · SDG 9

Carbon Footprint Predictor (SDG 13)

Regress CO_2 per passenger-km from route, load factor, aircraft type, and season. Feed BI sustainability dashboard.

Regression · SDG 13

NLP OBJECTIVE — Passenger Voice Mining

What We'll Build

1. Sentiment Analysis on open-text feedback and review datasets (positive / neutral / negative)
2. Topic Modelling (LDA / BERTopic) to auto-discover recurring complaint themes (e.g. legroom, delay, food)
3. Aspect-Based Sentiment: per service dimension (Wi-Fi, seat, boarding) from free text
4. Keyword extraction to surface trending issues in real-time for operations teams
5. LLM-assisted review summarisation — generate monthly insight reports automatically

Business Value

Voice of Customer Index

Real-time sentiment score per route

Complaint Heatmap

Topic × route matrix for ops teams

Churn Signal Detection

Negative language → alert for retention

NPS Augmentation

Enrich structured data with text signals

SDG 12 Reporting

Responsible service transparency metrics

COMPUTER VISION & DEEP LEARNING

Cabin Cleanliness Detection

CNN model to classify cabin photos (Clean / Needs Attention / Dirty). Automate pre-flight inspection scoring fed into satisfaction pipeline. SDG 12 alignment.

Airport Crowd Analytics

Object detection (YOLO) on CCTV feed images to count passengers and estimate queue length at boarding gates. Output feeds smart gate allocation (SDG 9).

Aircraft Condition Monitoring

CNN anomaly detection on aircraft exterior images to flag maintenance needs, reducing turnaround delays and improving operational efficiency.

Baggage Handling QC

Vision model to detect mishandled or damaged baggage at conveyor, automatically generating incident reports and improving SDG 12 responsible operations.

SUSTAINABLE DEVELOPMENT GOALS ALIGNMENT

SDG 9

- Smart Route Optimisation — predict load factor to maximise aircraft utilisation
- Airport Crowd Analytics — reduce gate congestion and boarding wait times
- Operational Efficiency Dashboard — real-time KPIs for airline ops managers
- Predictive maintenance signals to minimise ground delays

SDG 12

- Fuel usage per passenger-km — track and benchmark per route
- Cabin cleanliness CV model — reduce chemical waste in cleaning
- NLP transparency reports — responsible service quality communication
- Baggage handling QC — reduce damaged goods and waste

SDG 13

- Carbon footprint per passenger ML model — regress CO_2 by route/season
- Emission trend analysis BI dashboard — year-over-year carbon KPI
- Offset recommendation engine — flag high-emission routes
- Power BI sustainability scorecard integrated into executive reporting

GITHUB INTEGRATION & PROJECT MANAGEMENT

Repository Structure

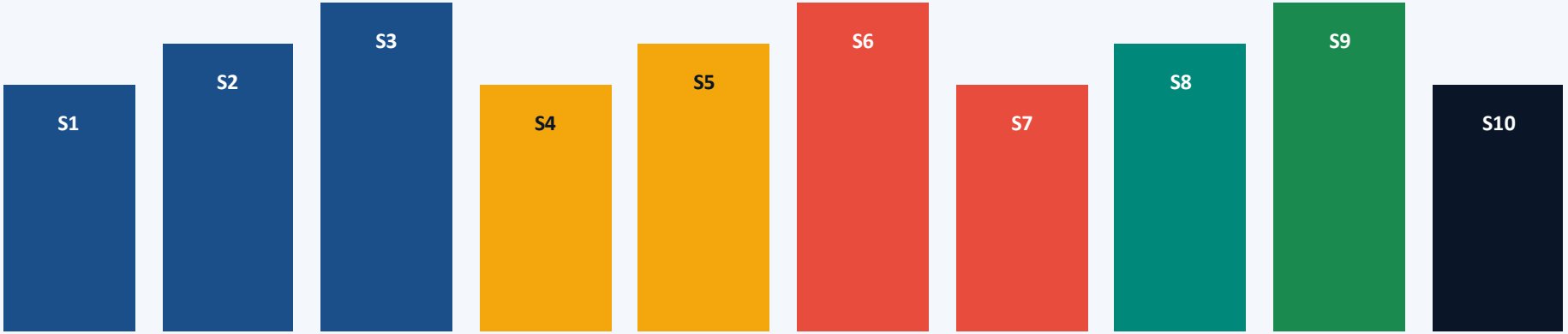
```
/data-warehouse   → SQL scripts, star schema, SSIS packages
/etl              → Python ETL pipelines (pandas, SQLAlchemy)
/bi-reports       → Power BI .pbix files + screenshots
/ml-models        → Jupyter notebooks, trained models (.pkl)
/nlp              → Text processing, LDA, BERTopic scripts
/computer-vision  → CNN training code, YOLO configs
/docs             → Architecture diagrams, report PDF
/dashboard        → Streamlit / Flask deployment
```

Agile Practices

- **Branching Strategy**
  main / develop / feature/* / hotfix branches
- **Pull Requests**
  Code review required for merges to develop
- **Issues & Milestones**
  GitHub Issues map to 10-sprint project plan
- **GitHub Actions CI**
  Automated linting + unit tests on push
- **README + Wiki**
  Architecture docs, setup guide, data dictionary
- **Releases**
  Tagged versions per sprint delivery milestone

PROJECT TIMELINE — 10 SPRINTS

Data Warehouse | Business Intelligence | Machine Learning | NLP | Computer Vision | Integration



S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
Kickoff & DWH Design	**ETL Pipeline**	**DWH Validation**	**Power BI – Core**	**Power BI – SDG**	**ML Models**	**ML – Routes & CO_2**	**NLP Pipeline**	**Computer Vision**	**Integration & Demo**
Star schema, data profiling	SSIS + Python ingestion	Data quality checks	Loyalty & satisfaction dashboards	Env. & ops dashboards	Classifier + churn predictor	Optimisation + carbon model	Sentiment + topic modelling	CNN cabin + crowd detection	GitHub, report, presentation

SKYINSIGHT

Turning flight data into sustainable, intelligent insights

BI | ML | NLP | Computer Vision | SDG 9 | 12 | 13 | GitHub